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Bigeneric initial ideals, diagonal subalgebras and bigraded Hilbert functions

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Abstract

In this paper we study some problems concerning bigraded ideals. By introducing the concept of bigeneric initial ideal, we answer an open question about diagonal subalgebras and we give a necessary condition for a function to be the bigraded Hilbert function of a bigraded algebra. Moreover, we give an upper bound for the regularity of a bistable ideal in terms of the degrees of its generators. © 2000 Elsevier Science B.V. All rights reserved.

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0. Introduction

In this work we introduce *bigeneric initial ideals* and apply them to different problems concerning bigraded algebras. More precisely, let k be an infinite field, let $R = k[x_1, \ldots, x_n, y_1, \ldots, y_m]$ be the polynomial ring in $n + m$ variables. We consider the grading defined by $\deg x_i = (1, 0)$, $\deg y_j = (0, 1)$. Let I be a bigraded ideal, i.e., an ideal which is homogeneous with respect to this grading.

Suppose that a monomial order is given on R. As in the graded case we make coordinate changes in order to obtain good properties for the initial ideal $\mathrm{in}(I)$ of I. However, in the bigraded situation the coordinate changes can only be made in the x's separately and in the y's separately. Roughly, the bigeneric initial ideal $\mathrm{bigin}(I)$ of an ideal I is defined to be the initial ideal of $g(I)$ where g is a general coordinate change

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in this restricted sense. The proof of the existence of bigeneric initial ideals is similar to the corresponding result for generic initial ideals. Furthermore, bigeneric initial ideals have some properties analogous to generic initial ideals. They are bi-Borel fixed. If k has characteristic zero, they are strongly bistable. In the first section we show these useful results and some other basic properties of bigeneric initial ideals.

We next study the Castelnuovo–Mumford regularity of bigeneric initial ideals. In the classical theory, the Bayer–Stillman Theorem [2] says that the regularity of a graded ideal equals the regularity of its generic initial ideal (with respect to a reverse lexicographical order). Example 2.4 shows that this is not true in general for a bigraded ideal I and bigin(I). We give an upper bound for the regularity of a bistable ideal in terms of the degrees of its generators (Theorem 2.2).

In Section 3 we apply bigeneric initial ideals to the study of diagonal subalgebras of a bigraded algebra. Recall that for a pair of positive integers a, b the (a, b)-diagonal Δ is the set $\Delta = \{(as, bs) \mid s \in \mathbb{Z}\}$. The diagonal subalgebra $(R/I)_\Delta$ of R/I is defined as $(R/I)_\Delta = \bigoplus_{s\in\mathbb{Z}} (R/I)_{(as,bs)}$. We show that there exist integers a', b' such that if $a \geq a'$, $b \geq b'$, then for the (a, b)-diagonal Δ we have that $(R/I)_\Delta$ has a quadratic Gröbner basis. This answers an open question in [6].

It is known that for any graded ideal there exists a unique lexsegment ideal with the same Hilbert function. In Section 4, introducing bilex ideals, we prove a similar result for bigraded ideals (Theorem 4.14). In contrast to the graded case, for a given bigraded ideal, a bilex ideal with the same bigraded Hilbert function is not unique (cf. Example 4.15). This suggests that it is hard to find a complete characterization of the bigraded Hilbert function as it is known in the graded case by Macaulay Theorem [11]. However, we give an upper bound for the growth of the bigraded Hilbert function of bigraded ideals (Theorem 4.18).

Bigraded algebras occur naturally in the context of Rees algebras (see [6]), which is one of the motivations for studying bigraded structures. However, notice that for all our results about bigraded objects, the corresponding multigraded results do hold as well with similar proofs.

1. Bigeneric initial ideals

Throughout the paper, let k be an infinite field, let $R = k[x_1, \ldots, x_n, y_1, \ldots, y_m]$ be the polynomial ring in $n + m$ variables. We consider the grading defined by $\deg x_i = (1, 0)$, $\deg y_j = (0, 1)$. An ideal $I \subset R$ is *bigraded* if it is homogeneous with respect to this grading. In this section we define bigeneric initial ideals and show some basic properties. First we need some notation.

Let $a, b \geq 0$ be integers. For a graded ideal $J \subset k[x_1, \ldots, x_n]$, the k-vector space generated by all elements of degree a is denoted by J_a. The ideal generated by J_a is denoted by (J_a). For a bigraded ideal $I \subset R$, we define $I_{(a,b)}$ and $(I_{(a,b)})$ analogously.

For a monomial $u \in R$, set

$$m_x(u) = \max\{i : x_i | u\}, \quad m_y(u) = \max\{i : y_i | u\}.$$

Recall that a monomial ideal $J \subset k[x_1,\dots,x_n]$ is *stable* if for every monomial $u \in J$ one has $x_i u/x_{m_x(u)} \in J$ for every $i < m_x(u)$. Furthermore, J is *strongly stable* if for every monomial $u \in J$ one has $x_i u/x_j \in J$ if u is divisible by x_j and $i < j$.

Definition 1.1. A monomial ideal $I \subset R$ is *bistable* if for every monomial $u \in I$ the following conditions are satisfied:

(1) for every $i < m_x(u)$ one has $x_i u/x_{m_x(u)} \in I$;
(2) for every $i < m_y(u)$ one has $y_i u/y_{m_y(u)} \in I$.

Definition 1.2. A monomial ideal $I \subset R$ is *strongly bistable* if for every monomial $u \in I$ the following conditions are satisfied:

(1) if u is divisible by x_j, then for every $i < j$ one has $x_i u/x_j \in I$;
(2) if u is divisible by y_l, then for every $i < l$ one has $y_i u/y_l \in I$.

Let $G = GL(n,k) \times GL(m,k)$ be the direct product of $GL(n,k)$ and $GL(m,k)$. An element $g = (d_{ij}, e_{kl}) \in G$ acts on the variables in R by

$$g : x_j \mapsto \sum_i d_{ij} x_i, \quad y_l \mapsto \sum_k e_{kl} y_k.$$

A monomial $\prod x_j^{a_j} \cdot \prod y_l^{b_l}$ is mapped by g to

$$\prod_j \left(\sum_i d_{ij} x_i \right)^{a_j} \cdot \prod_l \left(\sum_k e_{kl} y_k \right)^{b_l}.$$

Definition 1.3. A monomial ideal $I \subset R$ is *bi-Borel fixed* if $g(I) = I$ for any upper triangular matrix $g \in G$.

In the remainder of the section, assume that a monomial order is given on R which satisfies $x_1 > \cdots > x_n$, $y_1 > \cdots > y_m$. From standard topological arguments and following the proof of [7, Theorem 15.18], one has:

Theorem 1.4. *Let $I \subset R$ be a bigraded ideal. There is a Zariski open set U in G and an ideal J such that for all $g \in U$ we have* $\operatorname{in}(g(I)) = J$.

The ideal J in the theorem is defined to be the *bigeneric initial ideal* of I, denoted by $\operatorname{bigin}(I)$.

The proofs of the following two propositions are similar to the proofs of [7, Theorem 15.20] and [7, Theorem 15.23b], respectively.

Proposition 1.5. *The ideal* $\operatorname{bigin}(I)$ *is bi-Borel fixed for any bigraded ideal $I \subset R$.*

Proposition 1.6. *If k is a field of characteristic zero, then* $\operatorname{bigin}(I)$ *is strongly bistable for any bigraded ideal $I \subset R$.*

In Section 3 we shall use the following result.

Lemma 1.7. *Let $I \subset R$ be a bigraded ideal such that* $\mathrm{in}(I)$ *is bi-Borel fixed. Then there exist integers c', d' such that* $(\mathrm{in}(I)_{(c,d)})$ *is a bistable ideal for $c \geq c'$, $d \geq d'$.*

Proof. Put $J = \mathrm{in}(I)$. Let $(a_1, b_1), \ldots, (a_s, b_s)$ be the bidegrees of the monomials in the minimal set of generators of J. Set $a = \max_{1 \leq i \leq s}\{a_i\}$ and $b = \max_{1 \leq i \leq s}\{b_i\}$. Let

$$T = \{x^{\alpha_{ij}} y^{\beta_i} \colon 1 \leq i \leq r, 1 \leq j \leq s_i\}$$

be the minimal generating set for $(J_{(a,b)})$. Let $I_{(\beta_i)}$ be the ideal in $k[x_1, \ldots, x_n]$ generated by $\{x^{\alpha_{ij}} \colon 1 \leq j \leq s_i\}$. Let g be a direct product of an $n \times n$ upper triangular invertible matrix and the $m \times m$ identity matrix. If a monomial $u \in T$, then $g(u) \in T$, as well. Consequently, $I_{(\beta_i)}$ is Borel-fixed. Hence by [8, Proposition 10], there does exist an integer $c_i' \geq a$ such that $((I_{(\beta_i)})_r)$ is stable for all integers $r \geq c_i'$. Put $c' = \max\{c_i'\}$, and let d' be the corresponding integer for the y-part. If $x^\alpha y^\beta \in (J_{(c',d')})$, then $x^\alpha y^\beta = x^{\alpha'} y^{\beta'} \cdot x^{\alpha_{ij}} y^{\beta_i}$ for some i, j. Thus $x^\alpha \in ((I_{(\beta_i)})_{c_i'})$. It follows that $x_t x^\alpha / x_{m_x(x^\alpha)} \in ((I_{(\beta_i)})_{c_i'})$, if $t < m_x(x^\alpha)$. Thus $x_t x^\alpha y^\beta / x_{m_x}(x^\alpha) \in (J_{(c',d')})$. The analogous statement holds for the corresponding operations on y^β, whence indeed $(J_{(c',d')})$ is bistable. □

2. The regularity of bistable ideals

This section deals with bistable ideals in general (cf. Definition 1.1). In addition to previous notation, we denote by $M_{(b,d)}$ the set of all monomials of bidegree (b, d), and for a monomial u set

$$\min_x(u) = \min\{i \colon x_i | u\}, \quad \min_y(u) = \min\{i \colon y_i | u\}.$$

For convenience, we put $m_x(1) = m_y(1) = 0$ and $\min_x(1) = \min_y(1) = +\infty$.

Recall that the regularity of a bigraded R-module M is defined as follows:

$$\mathrm{reg}(M) = \max\{a - i \colon \mathrm{Tor}_i(M, k)_a \neq 0\}.$$

We define the x-regularity of M:

$$\mathrm{reg}_x(M) = \max\{b - i \colon \mathrm{Tor}_i(M, k)_{(b,d)} \neq 0\},$$

and similarly, the y-regularity:

$$\mathrm{reg}_y(M) = \max\{d - i \colon \mathrm{Tor}_i(M, k)_{(b,d)} \neq 0\}.$$

If $J \subset R$ is a monomial ideal, we write $G(J)$ for the unique minimal set of monomial generators of J.

It is clear that $I \subset R$ is a bistable ideal if and only if every element in $G(I)$ satisfies Definitions 1.1(1) and (2).

Similarly to the decomposition of monomials in stable ideals in the graded case (cf. [9, Lemma 1.1]), we have such a decomposition for bistable ideals which, however, is not unique.

Lemma 2.1. *Let I be a bistable ideal. For every monomial $v \in I$ there exists a decomposition $v = uw$ with $u \in G(I)$, $m_x(u) \leq \min_x(w)$ and $m_y(u) \leq \min_y(w)$.*

Proof. If $v \in I$ is a monomial, then $v = u'w'$ for some $u' \in G(I)$ and some monomial w'. Assume that $m_x(u') > i = \min_x(w')$. Then $v = (x_i u'/x_{m_x(u')})(x_{m_x(u')} w'/x_i)$ where, by Definition 1.1(1), $x_i u'/x_{m_x(u')} \in I$. Therefore $x_i u'/x_{m_x(u')} = u''z$ for some $u'' \in G(I)$, so that v can be written as $v = u''w''$ and $m_x(u'') \leq m_x(u')$. Now, if $m_x(u'') \leq \min_x(w'')$ we are done, otherwise we repeat the process. After finitely many steps, we get $v = uw$ with $u \in G(I)$ and $m_x(u) \leq \min_x(w)$. If $m_y(u) > j = \min_y(w)$, then writing $v = (y_j u/y_{m_y(u)})(y_{m_y(u)} w/y_j)$, we have that $\tilde{u} = y_j u/y_{m_y(u)} \in I$. As we have just shown, we can decompose $\tilde{u} = u_1 w_1$ with $u_1 \in G(I)$ and $m_x(u_1) \leq \min_x(w_1)$. Then $v = u_1(w_1 \tilde{w})$ where $\tilde{w} = (y_{m_y(u)} w/y_j)$, and $m_x(u_1) \leq \min_x(w_1) \leq m_x(\tilde{u}) = m_x(u) \leq \min_x(w) = \min_x(\tilde{w})$. Therefore $m_x(u_1) \leq \min_x(w_1 \tilde{w})$. Moreover, $m_y(u_1) \leq m_y(\tilde{u}) \leq m_y(u)$. After finitely many such steps, we obtain the desired decomposition of v. □

Theorem 2.2. *Let I be a bistable ideal, $G(I) = \{u_1, \ldots, u_s\}$, and $\deg u_i = (a_i, b_i)$ for $1 \leq i \leq s$. Let $m_x(I) = \max_{1 \leq i \leq s}\{a_i\}$ and $m_y(I) = \max_{1 \leq i \leq s}\{b_i\}$. Then*

(a) $\operatorname{reg}_x(I) = m_x(I)$ *and* $\operatorname{reg}_y(I) = m_y(I)$.

(b) $\operatorname{reg}(I) \leq m_x(I) + m_y(I)$.

First we show the theorem in a particular case.

Lemma 2.3. *Let $I \subset R$ be a bistable ideal such that $G(I) = G_x \bigcup G_y$ where $G_x \subset k[x_1, \ldots, x_n]$ and $G_y \subset k[y_1, \ldots, y_m]$. Then I satisfies (a) and (b) of Theorem* 2.2.

Proof. Let $F(G_x)$ and $F(G_y)$ be the minimal free resolutions of the ideals $(G_x) \subset R$ and $(G_y) \subset R$, respectively. Then it is known that $F(G_x) \otimes_R F(G_y)$ is the minimal free resolution of I. Since the ideals (G_x) and (G_y) are stable, by [9] (cf. also [1, Corollary 2.2]), $\operatorname{reg}(G_x) = m_x(G_x)$ and $\operatorname{reg}(G_y) = m_y(G_y)$. From here we conclude that Theorem 2.2 holds for I. □

Proof of Theorem 2.2. For a monomial $u = x^\alpha y^\beta \in M_{(a,b)}$, set:

$$|u| = a + b; \qquad u_x = x^\alpha, \quad u_y = y^\beta.$$

Here, we consider the degree lexicographical order on the monomials of R that satisfies $x_1 < x_2 < \cdots < x_n < y_1 < y_2 < \cdots < y_m$. Let $u_1 < u_2 < \cdots < u_s$.

First we show that for every $1 \leq t \leq s$, the ideal $I_t = (u_1, \ldots, u_t)$ is bistable. In fact, let $u \in G(I_t)$ and $i < m_x(u)$. Then $x_i u/x_{m_x(u)} \in I$, since I is bistable. If $x_i u/x_{m_x(u)} \in G(I)$, then as $x_i u/x_{m_x(u)} < u$, one has $x_i u/x_{m_x(u)} \in G(I_t)$. If $x_i u/x_{m_x(u)} = vw$ with $v \in G(I)$ and $w \neq 1$, then since $|v| < |x_i u/x_{m_x(u)}|$, one obtains that $v \in G(I_t)$. Therefore $x_i u/x_{m_x(u)} \in I_t$. Similarly one checks Definition 1.1(2).

We prove the theorem by induction on t. Consider the exact sequence of graded R-modules:

$$0 \to I_t \to I_{t+1} \to R/(I_t : v)(-|v|) \to 0 \tag{1}$$

where $v = u_{t+1}$.

We will show

(∗) $(I_t : v)$ is bistable with $G(I_t : v) = G_x \cup G_y$ where $G_x \subset k[x_1, \ldots, x_n]$ and $G_y \subset k[y_1, \ldots, y_m]$. Moreover, one has $m_x(I_t : v) \leq \max\{1, m_x(I_t) - |v_x|\}$ and $m_y(I_t : v) \leq \max\{1, m_y(I_t) - |v_y|\}$.

First, we claim that for all $i < m_x(v)$ and for all $j < m_y(v)$, one has $x_i \in (I_t : v)$ and $y_j \in (I_t : v)$. Indeed, let $i < m_x(v)$. Then $x_i v/x_{m_x(v)} \in I$, because I is bistable. If $x_i v/x_{m_x(v)} \in G(I)$, then from $x_i v/x_{m_x(v)} < v$ it follows that $x_i v/x_{m_x(v)} \in G(I_t)$, so that $x_i v \in I_t$. If $x_i v/x_{m_x(v)} = uw$ with $u \in G(I)$ and $w \neq 1$, then $|u| < |v|$, therefore one has $u < v$. Hence $u \in G(I_t)$, and $x_i v \in I_t$, as well. Similarly one shows that $(y_1, \ldots, y_{m_y(v)-1}) \subset (I_t : v)$.

Let now $z \in G(I_t : v) \setminus \{x_1, \ldots, x_{m_x(v)-1}, y_1, \ldots, y_{m_y(v)-1}\}$. It is then clear that $z \notin (x_1, \ldots, x_{m_x(v)-1}, y_1, \ldots, y_{m_y(v)-1})$, therefore $m_x(v) \leq \min_x(z)$ and $m_y(v) \leq \min_y(z)$. Since $zv \in I_t$ and I_t is bistable, we can apply Lemma 2.1 to get a decomposition $zv = uw$ with $u \in G(I_t)$ and $m_x(u) \leq \min_x(w)$, $m_y(u) \leq \min_y(w)$. Note that $w \neq 1$, because $u, v \in G(I)$. As z is a minimal generator of $(I_t : v)$, we have $z = u/\gcd(u, v)$ and $w = v/\gcd(u, v)$, where $\gcd(u, v)$ denotes the great common divisor of u and v.

Suppose first that $w_y \neq 1$. Then $z_y = 1$. In fact, assuming $z_y \neq 1$, we have

$$m_y(v) \leq \min_y(z) \leq m_y(u) \leq \min_y(w) \leq m_y(v),$$

therefore $\min_y(z) = \min_y(w)$ which contradicts to $\gcd(z, w) = 1$. Hence $z = z_x$. Since $z = u/\gcd(u, v)$, we get $u_y = \gcd(u_y, v_y)$, so that $z = u_x/\gcd(u_x, v_x)$ and $w = (v_x/\gcd(u_x, v_x))v_y/u_y$. Moreover one has $w_x = 1$. In fact, suppose $w_x \neq 1$. Then

$$m_x(v) \leq \min_x(z) \leq m_x(u) \leq \min_x(w) \leq m_x(v)$$

implies again the contradiction $\min_x(z) = \min_x(w)$. Hence $v_x = \gcd(u_x, v_x)$, so that $z = u_x/v_x$. Since $|u_x| \leq m_x(I_t)$, we get $|z| \leq m_x(I_t) - |v_x|$.

It remains to consider the case $w_y = 1$. Then $w = v_x/\gcd(u_x, v_x)$, and $v_y = \gcd(u_y, v_y)$. We show that $z_x = 1$. Indeed, supposing $z_x \neq 1$, one has

$$m_x(v) \leq \min_x(z) \leq m_x(u) \leq \min_x(w) \leq m_x(v),$$

therefore $\min_x(z) = \min_x(w)$ which contradicts to $\gcd(z, w) = 1$. Hence $z = u_y/v_y$ and we get $|z| \leq m_y(I_t) - |v_y|$.

It remains to show that $(I_t : v)$ is bistable. It is clear that every element in $\{x_1, \ldots, x_{m_x(v)-1}, y_1, \ldots, y_{m_y(v)-1}\}$ satisfies Definitions 1.1(1) and (2) Let $z \in G(I_t : v) \setminus \{x_1, \ldots, x_{m_x(v)-1}, y_1, \ldots, y_{m_y(v)-1}\}$, and $i < m_x(z)$. Then $m_x(z) \geq m_x(v)$, so that $m_x(zv) = m_x(z)$. Now, since $zv \in I_t$ and I_t is bistable, we get $(x_i z/x_{m_x(z)})v \in I_t$. Hence $x_i z/x_{m_x(z)} \in (I_t : v)$. Similarly, one checks Definition 1.1(2).

This concludes the proof of (∗).

The exact sequence (1) implies:

$$\operatorname{reg}(I_{t+1}) \leq \max\{\operatorname{reg}(I_t), \operatorname{reg}(R/(I_t : v)(-|v|))\}$$

(e.g. [7, Corollary 20.19]). Applying Lemma 2.3 to $(I_t : v)$ and using (∗), we obtain

$$\operatorname{reg}(I_t : v) \leq \max\{1, m_x(I_t) - |v_x|\} + \max\{1, m_y(I_t) - |v_y|\}.$$

By induction $\operatorname{reg} I_t \le m_x(I_t) + m_y(I_t) \le m_x(I_{t+1}) + m_y(I_{t+1})$. Moreover,

$$\begin{aligned}\operatorname{reg}(R/(I_t : v)(-|v|)) &= \operatorname{reg}(I_t : v) - 1 + |v| \\ &\le \max\{1, m_x(I_t) - |v_x|\} + \max\{1, m_y(I_t) - |v_y|\} - 1 \\ &\quad + |v_x| + |v_y|.\end{aligned}$$

By treating the different cases, since $|v_x| \le m_x(I_{t+1})$ and $|v_y| \le m_y(I_{t+1})$, we get

$$\operatorname{reg}(R/(I_t : v)(-|v|)) \le m_x(I_{t+1}) + m_y(I_{t+1})$$

and this proves (b) of the theorem.

To show (a), we consider the exact sequence

$$0 \to I_t \to I_{t+1} \to R/(I_t : v)(-|v_x|) \to 0. \tag{2}$$

From this sequence, we obtain

$$\operatorname{reg}_x(I_{t+1}) \le \max\{\operatorname{reg}_x(I_t), \operatorname{reg}_x(R/(I_t : v)(-|v_x|))\}.$$

By the inductive hypothesis $\operatorname{reg}_x(I_t) = m_x(I_t)$. Since the ideal $(I_t : v)$ satisfies $(*)$, by Lemma 2.3 we get $\operatorname{reg}_x(I_t : v) = m_x(I_t : v)$, so that

$$\operatorname{reg}_x(R/(I_t : v)(-|v_x|)) \le \max\{|v_x|, m_x(I_t) - 1\}.$$

As $|v_x| \le m_x(I_{t+1})$ and $m_x(I_t) \le m_x(I_{t+1})$, we get $\operatorname{reg}_x(I_{t+1}) \le m_x(I_{t+1})$. On the other hand, by definition, one has $m_x(I_{t+1}) \le \operatorname{reg}_x(I_{t+1})$. This concludes the proof of (a). □

It is known that a graded ideal and its generic initial ideal with respect to the reverse lexicographical order have the same regularity [2]. The corresponding result in the bigraded case does not hold:

Example 2.4. The following example was communicated to us by N. Terai. Let I be the ideal in $k[x_1, x_2, x_3, y_1, y_2, y_3]$ generated by

$$x_1x_2x_3, x_1x_2y_2, x_1x_3y_3, x_1y_1y_2, x_1y_1y_3,$$

$$x_2x_3y_1, x_2y_1y_3, x_2y_2y_3, x_3y_1y_2, x_3y_2y_3.$$

Then $\operatorname{reg}(I) = 3$. We consider the reverse lexicographical order with $x_1 > x_2 > x_3 > y_1 > y_2 > y_3$. Computing $\operatorname{bigin}(I)$ with Macaulay program, we obtain that $\operatorname{reg}(\operatorname{bigin}(I)) = 4$.

3. Gröbner bases for diagonal subalgebras

For a pair of positive integers a, b the (a, b)-diagonal Δ is the set $\Delta = \{(as, bs) | s \in \mathbb{Z}\}$. The *diagonal subalgebra* $(R/I)_\Delta$ of R/I is defined as $(R/I)_\Delta = \bigoplus_{s\in\mathbb{Z}} (R/I)_{(as,bs)}$. We consider $(R/I)_\Delta$ as a standard graded ring by assigning the degree one to the elements $x^\alpha y^\beta$ such that $|\alpha| = a$, $|\beta| = b$. In this section we shall show that for any bigraded ideal I in R, there exist a', b' such that whenever $a \ge a'$, $b \ge b'$, we have that the

(a,b)-diagonal subalgebra $(R/I)_\Delta$ of R/I has a quadratic Gröbner basis. This result is a bigraded correspondence to the well-known result that high enough Veronese subrings of any graded algebra have quadratic Gröbner bases (see [8]). Throughout the section, let $<_R$ be a monomial order on R satisfying $x_1 > \cdots > x_n$ and $y_1 > \cdots > y_m$. For some general results and terminology from the theory of Gröbner bases, see e.g. [10]. First recall some notation.

Let $<$ be an order on a polynomial ring $k[x_1,\ldots,x_n]$ and let A be a set of elements in $k[x_1,\ldots,x_n]$. The monomial x^α can be *reduced* with respect to A and $<$, if there exist a polynomial $f \in A$ and a monomial $x^\beta \in k[x_1,\ldots,x_n]$, such that $x^\alpha - fx^\beta = h$, where for the leading monomial $\mathrm{lm}(h)$ of h, we have $\mathrm{lm}(h) < x^\alpha$, or $h=0$. By abuse of notation we will write only $\mathrm{lm}(h) < x^\alpha$ for this. The monomial x^α is said to be *normal* with respect to A and $<$, if x^α cannot be reduced with respect to A and $<$. Notice that x^α is normal exactly if it cannot be written as $x^\alpha = x^{\alpha'}\,\mathrm{lm}(f)$, where $f \in A$, and $x^{\alpha'}$ is a monomial.

We shall use the following criteria for a set in an ideal to be a Gröbner basis.

Lemma 3.1. *Let $k|x_1,\ldots,x_n]$ be the polynomial ring, with an order $<$, and let $I \subset k[x_1,\ldots,x_n]$ be an ideal. Let $\mathscr{I}$ be a set of elements in the ideal I. Let N_I be the set of monomials which are normal with respect to I, and let $N_{\mathscr{I}}$ be the set of monomials which are normal with respect to $\mathscr{I}$. Then $N_I \subset N_{\mathscr{I}}$, and $\mathscr{I}$ is a Gröbner basis if and only if equality holds.*

Proof. If $\mathscr{I}$ is a Gröbner basis then clearly $N_I = N_{\mathscr{I}}$. Conversely, if $N_I = N_{\mathscr{I}}$ then for every monomial $x^\alpha \in \mathrm{in}(I)$, we have $x^\alpha \notin N_{\mathscr{I}}$. It follows that there exists $g \in \mathscr{I}$ such that x^α is divisible by $\mathrm{lm}(g)$, which implies that $\mathscr{I}$ is a Gröbner basis for I. □

For the (a,b)-diagonal Δ, we shall describe $(R/I)_\Delta$ as a quotient of a polynomial ring.

Let $T_{(a,b)}$ be the polynomial ring $T_{(a,b)} = K[\{z_u\}]$, where u is a monomial in R of bidegree (a,b). Let $\varphi = \varphi_{(a,b)} : T_{(a,b)} \to R$ be the homomorphism sending z_u to u. For a bigraded ideal $I \subset R$, let $V_{a,b}(I)$ denote the preimage of I in $T_{(a,b)}$ with respect to φ. Then we have $(R/I)_\Delta = T_{(a,b)}/V_{a,b}(I)$. Given the order $<_R$ on R, we consider an *extended order* $<$ on $T_{(a,b)}$. More precisely, let $<_{\mathrm{revlex}}$ be the reverse lexicographical order on $T_{(a,b)} = K[\{z_u\}]$ where the variables are ordered by $z_u < z_{u'}$ if $u <_R u'$. The extended order $<$ on $T_{(a,b)}$ is defined by $z^\alpha > z^\beta$ if $\varphi(z^\alpha) >_R \varphi(z^\beta)$ or if $\varphi(z^\alpha) = \varphi(z^\beta)$ and $z^\alpha >_{\mathrm{revlex}} z^\beta$.

Let σ be the k-linear map: $\mathrm{Im}\,\varphi \to T_{(a,b)}$ sending $x^\alpha y^\beta$ to the least monomial z^γ, with respect to the extended order, such that $\varphi(z^\gamma) = x^\alpha y^\beta$.

For $f \in R$, let $S(f)$ be the set $\{uf\}$, where u is a monomial of minimal bidegree such that $uf \in \mathrm{Im}\,\varphi$.

Remark 3.2. Note that if $f \in R$ is of bidegree (c,d), and if u is a minimal monomial such that $uf \in \mathrm{Im}\,\varphi$ in the above sense, then $\sigma(uf)$ is of degree $\max\{\lceil c/a\rceil, \lceil d/b\rceil\}$.

Essentially the same result as below is stated in [8, Theorem 18]. However, for the convenience of the reader we give a proof which is rather similar to the proof given in [8] for the graded case.

Theorem 3.3. *Let I be a bigraded ideal with Gröbner basis $g_1,\dots,g_r$ with respect to $<_R$. Assume that* $\mathrm{in}(I)$ *is bistable and that* $\deg(g_i)\le(c,d)$ *for* $1\le i\le r$. *If Δ is the (a,b)-diagonal, then the defining ideal $V_{(a,b)}(I)$ of $(R/I)_\Delta$ has a Gröbner basis with respect to the extended order consisting of the sets*

$$\{z_vz_{v'}-\sigma(\varphi(z_vz_{v'}))\}\quad\text{and}\quad\left\{\sigma(g)\colon g\in\bigcup_{i=1}^{r}S(g_i)\right\}.$$

In particular, the Gröbner basis consists of elements of degree $\le\max\{2,\lceil c/a\rceil,\lceil d/b\rceil\}$.

First we need some preliminary results. Notice that $\operatorname{Ker}\varphi$ is simply the defining ideal of the Segre product of two Veronese subrings of polynomial rings.

Lemma 3.4. *The set $\{z_vz_{v'}-\sigma(\varphi(z_vz_{v'}))\}$ is a Gröbner basis for* $\operatorname{Ker}\varphi$ *with respect to the extended order on $T_{(a,b)}$.*

Proof. Since $T_{(a,b)}/\operatorname{Ker}\varphi$ is a subring of R generated by monomials, it is well-known that $\operatorname{Ker}\varphi$ is generated by the set $\{z^\alpha-z^\beta\colon\varphi(z^\alpha)=\varphi(z^\beta)\}$. Hence for any monomial z^γ, the normal form of z^γ with respect to $\operatorname{Ker}\varphi$ equals $\sigma(\varphi(z^\gamma))$. Write $\sigma(\varphi(z^\gamma))$ as $\sigma(\varphi(z^\gamma))=z_{v_1}\cdots z_{v_q}$, where $z_{v_1}\ge\cdots\ge z_{v_q}$. One verifies that $\sigma(\varphi(z^\gamma))$ is the unique monomial with the same image as z^γ satisfying the following condition: If x_i, y_j are variables in v_l, and $x_{i'}$, $y_{j'}$ are variables in v_{l+1}, then $i\le i',j\le j'$. Set $z^\gamma=z_{w_1}\cdots z_{w_q}$, where $z_{w_1}\ge\cdots\ge z_{w_q}$. If z^γ is not normal with respect to $\operatorname{Ker}\varphi$, then there exist $1\le i<j\le q$ such that x_t is a variable in w_i and x_s is a variable in w_j, where $t>s$, or similarly for two y-variables. Then $z_{w_i}z_{w_j}\ne\sigma(\varphi(z_{w_i}z_{w_j}))$, and hence z^γ is not normal with respect to the claimed Gröbner basis. We are done by Lemma 3.1. □

The following properties of the extended order $<$ are obvious:

Remark 3.5. (1) If z^α, z^β are monomials such that $z^\alpha=\sigma(\varphi(z^\alpha))$ and $z^\alpha>z^\beta$, then $\varphi(z^\alpha)>\varphi(z^\beta)$;

(2) For every polynomial $f\in\operatorname{Im}\varphi$ we have $\operatorname{lm}(\sigma(f))=\sigma(\operatorname{lm}(f))$.

Lemma 3.6. *The set consisting of $\{z_vz_{v'}-\sigma(\varphi(z_vz_{v'}))\}$ and of all elements of the form $\sigma(ug_i)$, $1\le i\le r$, where u is a monomial such that $ug_i\in\operatorname{Im}\varphi$, is a Gröbner basis for $V_{a,b}(I)$ with respect to the extended order.*

Proof. Let z^γ be a monomial which is not normal with respect to the ideal $V_{a,b}(I)$. By Lemma 3.1 it is enough to show that z^γ is also not normal with respect to the claimed Gröbner basis. By assumption, there exists some polynomial $f\in V_{a,b}(I)$ such that

$z^\gamma > \operatorname{lm}(z^\gamma - f)$. If z^γ can be reduced by elements of the form $z_v z_{v'} - \sigma(\varphi(z_v z_{v'}))$, we are done. Otherwise z^γ is normal with respect to $\operatorname{Ker}\varphi$ by Lemma 3.4, which gives $z^\gamma = \sigma(\varphi(z^\gamma))$. By Remark 3.5(1), it follows that $\varphi(z^\gamma) > \varphi(\operatorname{lm}(z^\gamma - f)) \geq \operatorname{lm}(\varphi(z^\gamma) - \varphi(f))$. This means that $\varphi(z^\gamma)$ is not normal with respect to I, and consequently $\varphi(z^\gamma) = u \operatorname{lm}(g_i)$ for some monomial u and some g_i. Thus $z^\gamma = \sigma(\varphi(z^\gamma)) = \sigma(\operatorname{lm}(ug_i)) = \operatorname{lm}(\sigma(ug_i))$. Consequently z^γ is not normal with respect to the claimed Gröbner basis, which completes the proof. □

Proof of Theorem 3.3. By Lemma 3.6 we are done if for any element of the form $\sigma(wg_i)$, where w is a monomial such that $wg_i \in \operatorname{Im}\varphi$, we have that $\operatorname{lm}(\sigma(wg_i))$ is divided by $\operatorname{lm}(\sigma(w'g_j))$, where $w'g_j \in S(g_j)$. Thus if $\tilde{u} \in \operatorname{in}(I) \cap \operatorname{Im}\varphi$, we need that $\sigma(\tilde{u})$ is divided by some $\sigma(u)$ where u is a minimal generator for $\operatorname{in}(I) \cap \operatorname{Im}\varphi$. Let u' be an element in $\operatorname{in}(I) \cap \operatorname{Im}\varphi$ of minimal bidegree among those dividing $\tilde{u}$. Say $\deg(u') = t(a,b)$. Let $\tilde{u} = x_{i_1} \cdots x_{i_{d_1}} \cdot y_{j_1} \cdots y_{j_{d_2}}$ where $i_1 \leq \cdots \leq i_{d_1}$ and $j_1 \leq \cdots \leq j_{d_2}$. Since $\operatorname{in}(I)$ is bistable we have that $u = x_{i_1} \cdots x_{i_{ta}} \cdot y_{j_1} \cdots y_{j_{tb}} \in \operatorname{in}(I)$. Furthermore, u is a minimal generator for $\operatorname{in}(I) \cap \operatorname{Im}\varphi$ since it has bidegree $t(a,b)$. As $\sigma(u)$ divides $\sigma(\tilde{u})$ we are done. □

Lemma 3.7. *Let $I \subset R$ be a bigraded ideal with Gröbner basis of bidegree $\leq (c,d)$, and* let $a \geq c, b \geq d$. *Then*

(1) $(I_{(a,b)}) = ((I_{(c,d)})_{(a,b)}) = \{g \in I\colon \deg g \geq (a,b)\}$;

(2) $((\operatorname{in}(I))_{(a,b)}) = \operatorname{in}((I_{(a,b)}))$;

(3) $(I_{(a,b)})$ *has a Gröbner basis consisting of elements which are all of bidegree* (a,b).

Proof. The first claim is true since I is generated in bidegree $\leq (c,d)$. Next, if the set $\{f_i\}$ is a Gröbner basis for I, then $\operatorname{in}(I)$ is generated by the set $\{\operatorname{lm}(f_i)\}$. Thus, $\operatorname{in}((I_{(a,b)}))$ is generated by all elements of the form $u\operatorname{lm}(f_i)$ where u is a monomial such that $\deg u \operatorname{lm}(f_i) = (a,b)$. Clearly $((\operatorname{in}(I))_{(a,b)})$ is generated by the same set and thus $((\operatorname{in}(I))_{(a,b)}) = \operatorname{in}((I_{(a,b)}))$. It follows that the set of elements of the form uf_i where u is a monomial such that $\deg u\operatorname{lm}(f_i) = (a,b)$ is a Gröbner basis for $(I_{(a,b)})$. □

Corollary 3.8. *Let $I \subset R$ be a bigraded ideal such that* $\operatorname{in}((I_{(c,d)}))$ *is bistable for some c,d, where* $\operatorname{in}(I)$ *is generated in bidegree $\leq (c,d)$. Let $\{g_i\}$ be a Gröbner basis for $I_{(c,d)}$, where* $\deg(g_i) = (c,d)$ *for all i (see Lemma* 3.7). *Assume that $a \geq c, b \geq d$. If Δ is the (a,b)-diagonal, then the defining ideal $V_{(a,b)}(I)$ of $(R/I)_\Delta$ has a Gröbner basis consisting of the sets $\{z_u z_{u'} - \sigma(\varphi(z_u z_{u'}))\}$ and $\{\sigma(g)\}$ where g is an element the form vg_i, where v is a monomial of bidegree $(a-c, b-d)$. In particular, the Gröbner basis for $V_{(a,b)}(I)$ consists of quadratic elements.*

Proof. Put $J = (I_{(c,d)})$. By Lemma 3.7 $\operatorname{in}(J) = \operatorname{in}(I_{(c,d)}) = ((\operatorname{in} I)_{(c,d)})$. Thus $\operatorname{in}(J)$ is bistable. By the same lemma we have $(I_{(a,b)}) = (J_{(a,b)})$. Since $\operatorname{Im}\varphi$ consists of elements

of bidegree $\geq (a,b)$, it follows that

$$V_{(a,b)}(I) = V_{(a,b)}((I_{(a,b)})) = V_{(a,b)}((J_{(a,b)})) = V_{(a,b)}(J).$$

We are done since Theorem 3.3 applies to J. □

Theorem 3.9. *For any bigraded ideal $I \subset R$, there exist a', b' such that for all (a,b)-diagonals, where $a \geq a'$, $b \geq b'$, the diagonal subalgebra $(R/I)_\Delta$ of R/I has a quadratic Gröbner basis. If k has characteristic zero, and if* bigin(I) *is generated in bidegree $\leq (c,d)$, then one may choose $a' = \lceil c/2 \rceil$, $b' = \lceil d/2 \rceil$.*

Proof. By Proposition 1.5, for any bigraded ideal $I \subset R$, we have that bigin(I) is bi-Borel fixed. After a coordinate change we may thus assume that in(I) is bi-Borel fixed. In the case when k has characteristic zero, in(I) is bistable by Proposition 1.6, and the result follows from Theorem 3.3. In the general case, there exist integers c', d' such that in(I) is generated in bidegree $\leq (c', d')$, and such that $(\text{in}(I)_{(c,d)})$ is bistable whenever $c \geq c', d \geq d'$ by Lemma 1.7. Now the previous corollary applies. □

4. The bigraded Hilbert function

In this section we apply bigeneric initial ideals to study the bigraded Hilbert function of bigraded algebras. We take as a model the proof of Macaulay Theorem [11] as it is given in [4]. Our main results here are Theorems 4.14 and 4.18 below. Example 4.15 shows that the situation is different from the graded case where a lexsegment ideal I^{lex} with the same Hilbert function as I is unique.

Let $I \subset R$ be a bigraded ideal. Set $h(b,d) = \dim_k R_{(b,d)}/I_{(b,d)}$ where $R_{(b,d)}$ denotes the vector space of all bihomogeneous elements in R of bidegree (b,d), and $I_{(b,d)} = I \cap R_{(b,d)}$. We consider the bigraded Hilbert function of R/I:

$$H(R/I) = \sum_{b,d \geq 0} h(b,d) t^b s^d.$$

In addition to the notation introduced at the beginning of Section 2, we set

$$X = \{x_1, \ldots, x_n\}, \qquad Y = \{y_1, \ldots, y_m\},$$

and for a subset $S \subset M_{(b,d)}$, denote

$$m_{x,i}(S) = |\{u \in S : m_x(u) = i\}|, \qquad m_{y,i}(S) = |\{u \in S : m_y(u) = i\}|,$$

$$m_{x,\leq i}(S) = \sum_{j=1}^{i} m_{x,j}(S), \qquad m_{y,\leq i}(S) = \sum_{j=1}^{i} m_{y,j}(S).$$

Throughout this section we will work with the degree lexicographical order on R induced by $x_1 > x_2 > \cdots > x_n > y_1 > y_2 > \cdots > y_m$. However, all results hold if we consider any term order which is the degree lexicographical order on $k[x_1, \ldots, x_n]$ with $x_1 > x_2 > \cdots > x_n$ and the degree lexicographical order on $k[y_1, \ldots, y_m]$ with $y_1 > y_2 > \cdots > y_m$ (see also Remark 4.7 below).

Definition 4.1. For a monomial $u \in M_{(b,d)}$, the set $L(u)=\{w \in M_{(b,d)}: w \geq u\}$ is called a *lexsegment*.

Remark 4.2. First note that for a given integer $\ell \leq |M_{(b,d)}|$, there exists a unique lexsegment $L \subset M_{(b,d)}$ with $|L| = \ell$.

If $u \in M_{(b,0)}$, then $L(u)$ is the usual lexsegment in $M_{(b,0)}$, as defined in the homogeneous case. In this case, it is clear that the shadow of $L(u)$, defined by $XL(u) = \{x_i w, 1 \leq i \leq n, w \in L(u)\}$, is again a lexsegment, namely $XL(u) = L(ux_n)$.

Let $S \subset M_{(b,0)}$ be a strongly stable set, and let S^{lex} be the lexsegment in $M_{(b,0)}$ with $|S^{\text{lex}}| = |S|$. Then Bigatti [4] proved:

(a) $m_{x,\leq i}(S^{\text{lex}}) \leq m_{x,\leq i}(S)$ for all i.

(b) $|XS^{\text{lex}}| \leq |XS|$.

Lemma 4.3. *Let S be a strongly bistable set of monomials in $M_{(b,d)}$, and let I be the ideal generated by S. Then*

(a) $XS = \bigcup_{u \in S}\{ux_{m_x(u)}, ux_{m_x(u)+1}, \ldots, ux_n\}$,

$$YS = \bigcup_{u \in S}\{uy_{m_y(u)}, uy_{m_y(u)+1}, \ldots, uy_m\},$$

and these unions are disjoint.

(b) $m_{x,i}(XS) = m_{x,\leq i}(S)$, *and* $m_{y,i}(YS) = m_{y,\leq i}(S)$.

(c) $|XS| = \sum_{i=1}^{n} m_{x,\leq i}(S)$, and $|YS| = \sum_{i=1}^{m} m_{y,\leq i}(S)$.

(d) $\dim_k I_{(b+1,d)} = \sum_{i=1}^{n} m_{x,\leq i}(S)$, *and* $\dim_k I_{(b,d+1)} = \sum_{i=1}^{m} m_{y,\leq i}(S)$.

Proof. (a) Let $v \in XS$. Then $v = ux_i$ with $u \in S$ and some $1 \leq i \leq n$. If $i < m_x(u)$, then $v = (x_i u / x_{m_x(u)})x_{m_x(u)}$, and since S is bistable, one has $u' = x_i u / x_{m_x(u)} \in S$. Therefore $v = u' x_{m_x(u)}$ with $m_x(u) \geq m_x(u')$ as desired. To see that this decomposition is unique, assume that $v = u_1 x_j = u_2 x_q$ with $u_1, u_2 \in S$ and $j \geq m_x(u_1)$, $q \geq m_x(u_2)$. Then $j = m_x(u_1 x_j) = m_x(u_2 x_q) = q$, so that $u_1 = u_2$. This shows (a).

It is clear that one has the implications: (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d). $\square$

Definition 4.4. A subset of monomials $L \subset M_{(b,d)}$ is called *bilex* if for every monomial $x^\alpha y^\beta \in L$ the following conditions are satisfied:

(xlex) if $u \in M_{(b,0)}$ and $u > x^\alpha$, then $uy^\beta \in L$;

(ylex) if $v \in M_{(0,d)}$ and $v > y^\beta$, then $x^\alpha v \in L$.

A monomial ideal $J \subset R$ is called a *bilex ideal* if $J_{(b,d)}$ is generated, as a k-vector space, by a bilex set of monomials for every $b, d \geq 0$.

Note that bilex ideals are strongly bistable.

Lemma 4.5. *If $L \subset M_{(b,d)}$ is a bilex set, then XL and YL are bilex too.*

Proof. Take a monomial $uv \in L$ with $u \in M_{(b,0)}$, $v \in M_{(0,d)}$, and let $w > x_i u$. Then $w > x_n u$, so that $w \in L(x_n u) = XL(u)$. Therefore $w = x_j z$ for some $z \in L(u)$. Since L

is bilex, one has $zv \in L$, hence $wv \in XL$. Similarly, one shows that XL satisfies (ylex). By symmetry, YL is bilex too. □

Lemma 4.6. *Let L be a lexsegment in $M_{(b,d)}$. Then*
(a) *L is bilex.*
(b) *YL is a lexsegment.*

Proof. One checks immediately (a). To prove (b), say $L = L(u)$ where $u \in M_{(b,d)}$. We will show that $YL(u) = L(uy_m)$. We may assume $d > 0$. If $v \in L(u)$, then for every $1 \leq j \leq m$, one has $y_j v \geq y_j u \geq y_m u$, therefore $YL(u) \subset L(uy_m)$.

Let now $w \in L(uy_m)$. Write $u = u_x u_y$ and $w = w_x w_y$ with $u_x, w_x \in M_{(b,0)}$ and $u_y \in M_{(0,d)}$, $w_y \in Y^{d+1}$. Since $w \geq uy_m$, one has $w_x \geq u_x$. If $w_x > u_x$, then for any y_i dividing w_y, one has $w_x w_y / y_i > u_x u_y$, therefore $w/y_i \in L(u)$ which implies that $w \in YL(u)$. It remains to consider the case $w_x = u_x$. Then $w_y \geq u_y y_m$, so that $w_y \in L(u_y y_m) = YL(u_y)$. Therefore $w_y = y_j v$ for some $v \in L(u_y)$ and some j. Then $u_x v \geq u_x u_y$, and since $w = y_j u_x v$ we obtain that $w \in YL(u)$. □

Remark 4.7. (a) Notice that in general XL is not a lexsegment.

Consider the following simple example: $R = k[x_1, x_2, x_3, y_1, y_2]$, $u = x_2 y_1$, and $L = L(u) = \{x_1 y_1, x_1 y_2, x_2 y_1\}$. Then $x_2^2 y_2 > x_2 x_3 y_1$, so that $x_2^2 y_2 \in L(ux_3)$, but $x_2^2 y_2 \notin XL(u)$.

(b) If we consider the degree lexicographical order with $y_1 > \cdots > y_m > x_1 > \cdots > x_n$, we obtain that XL is a lexsegment.

Let S be a strongly bistable set of monomials in $M_{(b,d)}$. Set $T = \{v \in M_{(0,d)} : uv \in S \text{ for some } u \in M_{(b,0)}\}$ and let $T = \{v_1, \ldots, v_k\}$ where $v_1 > v_2 > \cdots > v_k$. Then S can be decomposed uniquely as a disjoint union

$$S = \bigcup_{j=1}^{k} S_j v_j \quad \text{with } S_j \subset M_{(b,0)} \text{ for } 1 \leq j \leq k. \tag{3}$$

For each j, $1 \leq j \leq k$, let S_j^{lex} be the lexsegment in $M_{(b,0)}$ with $|S_j^{\text{lex}}| = |S_j|$. Set

$$S^{\text{xlex}} = \bigcup_{j=1}^{k} S_j^{\text{lex}} v_j.$$

By symmetry, one can decompose S as a disjoint union

$$S = \bigcup_{j=1}^{l} u_j T_j \quad \text{with } T_j \subset M_{(0,d)} \text{ for } 1 \leq j \leq l, \tag{4}$$

where $\{u_1, \ldots, u_l\} = \{u \in M_{(b,0)} : uv \in S \text{ for some } v \in M_{(0,d)}\}$ and $u_1 > u_2 > \cdots > u_l$. For each j, $1 \leq j \leq l$, let T_j^{lex} be the lexsegment in $M_{(0,d)}$ with $|T_j^{\text{lex}}| = |T_j|$. Set

$$S^{\text{ylex}} = \bigcup_{j=1}^{l} u_j T_j^{\text{lex}}.$$

Theorem 4.8. *Let S be a strongly bistable set of monomials in $M_{(b,d)}$. Then there exists a bilex set $S^{\mathrm{bilex}} \subset M_{(b,d)}$ such that*

(a) $|S^{\mathrm{bilex}}| = |S|$;

(b) $m_{x,\leq i}(S^{\mathrm{bilex}}) \leq m_{x,\leq i}(S)$;

(c) $m_{y,\leq i}(S^{\mathrm{bilex}}) \leq m_{y,\leq i}(S)$.

Proof. Let $S = \bigcup_{j=1}^{k} S_j v_j$ be decomposition (3) of S. Since S is strongly bistable, we have that $T = \{v_1, \ldots, v_k\}$ is a strongly stable subset of $M_{(0,d)}$, and for every j the set S_j is strongly stable in $M_{(b,0)}$. We will show that the set S^{xlex} defined above has the following properties:

$(*)$ if $w = uv_j \in S^{\mathrm{xlex}}$ where $u \in S_j^{\mathrm{lex}}$, then $L(u)v_j \subset S^{\mathrm{xlex}}$;

$(**)$ if $w = uv_j \in S^{\mathrm{xlex}}$ and y_s divides v_j, then for every $l < s$ one has $uy_l v_j / y_s \in S^{\mathrm{xlex}}$.

In fact, S_j^{lex} being a lexsegment, one has $L(u) \subset S_j^{\mathrm{lex}}$, and this implies $(*)$. To prove $(**)$, note that since T is strongly stable, one has $y_l v_j / y_s \in T$, so that $y_l v_j / y_s = v_q$ for some $1 \leq q < j$. We have to show that $u \in S_q^{\mathrm{lex}}$. The set S being strongly bistable, for every $z \in S_j$ one has $zv_q \in S$, so that $z \in S_q$. Therefore $S_j \subset S_q$ which implies the inclusion $S_j^{\mathrm{lex}} \subset S_q^{\mathrm{lex}}$. This shows that $u \in S_q^{\mathrm{lex}}$. Hence S^{xlex} satisfies $(*)$ and $(**)$, so that, in particular, S^{xlex} is a strongly bistable set, too.

Let now $S^{\mathrm{xlex}} = \bigcup_{j=1}^{l} u_j T_j$ be the decomposition (4) of S^{xlex}. Since S^{xlex} satisfies $(*)$, one obtains that $U = \{u_1, \ldots, u_l\}$ is a lexsegment in $M_{(b,0)}$. From property $(**)$ it follows that each T_j is strongly stable in $M_{(0,d)}$. Set

$$S^{\mathrm{bilex}} = (S^{\mathrm{xlex}})^{\mathrm{ylex}}.$$

Note that by construction $S^{\mathrm{bilex}} = \bigcup_{j=1}^{l} u_j T_j^{\mathrm{lex}}$. We will show that S^{bilex} is bilex, and this proves (a). Let $w \in S^{\mathrm{bilex}}$. Then $w = u_j v$ with $v \in T_j^{\mathrm{lex}}$. Since T_j^{lex} is a lexsegment, it is clear that $u_j L(v) \subset S^{\mathrm{bilex}}$. Now, let $u \in L(u_j)$. Since U is a lexsegment and $u_j \in U$, one has that $u \in U$, so that $u = u_s$ for some $s \leq j$. We have to show that $v \in T_s^{\mathrm{lex}}$. Since S^{xlex} satisfies $(*)$, one obtains, as above, that $T_j \subset T_s$. Therefore $T_j^{\mathrm{lex}} \subset T_s^{\mathrm{lex}}$, and we are done.

To prove (b) and (c) note that by Remark 4.2(b) one has $m_{x,\leq i}(S_j^{\mathrm{lex}}) \leq m_{x,\leq i}(S_j)$ for every $1 \leq j \leq k$, and $m_{y,\leq i}(T_j^{\mathrm{lex}}) \leq m_{y,\leq i}(T_j)$ for every $1 \leq j \leq l$. It follows easily that for every i

$$m_{x,\leq i}(S^{\mathrm{bilex}}) = m_{x,\leq i}(S^{\mathrm{xlex}}) = \sum_{j=1}^{k} m_{x,\leq i}(S_j^{\mathrm{lex}}) \leq \sum_{j=1}^{k} m_{x,\leq i}(S_j) = m_{x,\leq i}(S)$$

and

$$m_{y,\leq i}(S^{\mathrm{bilex}}) = \sum_{j=1}^{l} m_{y,\leq i}(T_j^{\mathrm{lex}}) \leq \sum_{j=1}^{l} m_{y,\leq i}(T_j) = m_{y,\leq i}(S^{\mathrm{xlex}}) = m_{y,\leq i}(S). \qquad \square$$

It follows that:

Corollary 4.9. *Let I be a strongly bistable ideal generated in degree (b,d). Then there exists a bilex ideal J generated in degree (b,d) such that*

(a) $\dim_k J_{(b,d)} = \dim_k I_{(b,d)}$;

(b) $\dim_k J_{(b+1,d)} \leq \dim_k I_{(b+1,d)}$;
(c) $\dim_k J_{(b,d+1)} \leq \dim_k I_{(b,d+1)}$.

Proof. Let S be the minimal set of monomial generators of I, and let J be the ideal generated by the set S^{bilex} defined in Theorem 4.8. Then by Lemma 4.5, J is a bilex ideal, and by Lemma 4.3(d) satisfies (b) and (c). □

A bilex set is not defined uniquely by the properties (a)–(c) of Theorem 4.8. We can take $S^{\text{bilex}_2} = (S^{\text{ylex}})^{\text{xlex}}$ which is bilex too, and satisfies (a)–(c) of Theorem 4.8. The following example shows that the bilex sets can be different.

Example 4.10. Let $R = k[x_1, x_2, x_3, y_1, y_2, y_3]$ and

$$S = \{x_1^2y_1^2, x_1^2y_1y_2, x_1^2y_1y_3, x_1^2y_2^2, x_1x_2y_1^2, x_1x_2y_1y_2, x_1x_2y_1y_3, x_1x_2y_2^2, x_1x_3y_1^2,$$
$$x_1x_3y_1y_2, x_1x_3y_2^2, x_2^2y_1^2, x_2^2y_1y_2, x_2^2y_1y_3\}.$$

Then S is a strongly bistable set, and

$$S^{\text{xlex}} = \{x_1^2y_1^2, x_1^2y_1y_2, x_1^2y_1y_3, x_1^2y_2^2, x_1x_2y_1^2, x_1x_2y_1y_2, x_1x_2y_1y_3, x_1x_2y_2^2, x_1x_3y_1^2,$$
$$x_1x_3y_1y_2, x_1x_3y_1y_3, x_1x_3y_2^2, x_2^2y_1^2, x_2^2y_1y_2\}$$

is bilex, i.e. $S^{\text{bilex}} = S^{\text{xlex}}$. On the other hand, the set

$$S^{\text{ylex}} = \{x_1^2y_1^2, x_1^2y_1y_2, x_1^2y_1y_3, x_1^2y_2^2, x_1x_2y_1^2, x_1x_2y_1y_2, x_1x_2y_1y_3, x_1x_2y_2^2, x_1x_3y_1^2,$$
$$x_1x_3y_1y_2, x_1x_3y_1y_3, x_2^2y_1^2, x_2^2y_1y_2, x_2^2y_1y_3\}$$

is also bilex and $S^{\text{bilex}_2} = S^{\text{ylex}} \neq S^{\text{bilex}}$.

Lemma 4.11. *Let $S \subset T$ be strongly bistable sets in $M_{(b,d)}$. Then*
(a) $S^{\text{xlex}} \subset T^{\text{xlex}}$;
(b) $S^{\text{ylex}} \subset T^{\text{ylex}}$;
(c) $S^{\text{bilex}} \subset T^{\text{bilex}}$.

Proof. Decomposing $S = \bigcup_{i=1}^{k} S_i v_i$ and $T = \bigcup_{i=1}^{s} T_i w_i$ according to (3), since $S \subset T$ one has that $\{v_1, \ldots, v_k\}$ is a subset of $\{w_1, \ldots, w_s\}$. Therefore for every i, $v_i = w_{j(i)}$ and $S_i w_{j(i)} \subset T$ which implies that $S_i \subset T_{j(i)}$. Thus $S_i^{\text{lex}} \subset T_{j(i)}^{\text{lex}}$, so that $S_i^{\text{lex}} v_i \subset T_{j(i)}^{\text{lex}} w_{j(i)} \subset T^{\text{xlex}}$ for each i. Hence $S^{\text{xlex}} \subset T^{\text{xlex}}$.

Similarly one shows (b). Finally, (c) follows from (a) and (b). □

Lemma 4.12. *Let $S \subset M_{(b,d)}$ be a strongly bistable set. Then $XS^{\text{bilex}} \subset (XS)^{\text{bilex}}$ and $YS^{\text{bilex}} \subset (YS)^{\text{bilex}}$.*

Proof. First we show the inclusion $XS^{\text{xlex}} \subset (XS)^{\text{xlex}}$. Let $S = \bigcup_{i=1}^{k} S_i v_i$ be decomposition (3) of S. Then one has $XS = \bigcup_{i=1}^{k} XS_i v_i$, so that $(XS)^{\text{xlex}} = \bigcup_{i=1}^{k} (XS_i)^{\text{lex}} v_i$. On the other hand, $XS^{\text{xlex}} = \bigcup_{i=1}^{k} XS_i^{\text{lex}} v_i$. By construction we have that $|(XS_i)^{\text{lex}}| = |XS_i|$. Since $|XS_i| \geq |XS_i^{\text{lex}}|$ (Remark 4.2(b)), and $(XS_i)^{\text{lex}}$ and XS_i^{lex} are both lexsegments of the same degree, one obtains that $XS_i^{\text{lex}} \subset (XS_i)^{\text{lex}}$. Thus $XS^{\text{xlex}} \subset (XS)^{\text{xlex}}$.

We claim that $S_i^{\text{lex}} \subset S_1^{\text{lex}}$ for every i. In fact, since y_1^d is contained in every strongly stable set in $M_{(0,d)}$, and since S^{xlex} satisfies $(**)$, one obtains that $v_1 = y_1^d$ and $S_i^{\text{lex}} \subset S_1^{\text{lex}}$.

Let $S_1^{\text{lex}} = \{u_1, \ldots, u_l\}$ with $u_1 > \cdots > u_l$. Then $S^{\text{xlex}} = \bigcup_{i=1}^{l} u_i T_i$ with $T_1 = \{v_1, \ldots, v_k\}$, and $S^{\text{bilex}} = \bigcup_{i=1}^{l} u_i T_i^{\text{lex}}$. Let $(XS_1)^{\text{lex}} = \{w_1, \ldots, w_s\}$. Then we get the decomposition $(XS)^{\text{xlex}} = \bigcup_{i=1}^{s} w_i W_i$ with $W_i \subset M_{(0,d)}$, so that $(XS)^{\text{bilex}} = \bigcup_{i=1}^{s} w_i W_i^{\text{lex}}$. Fix an i and take a $v \in T_i$. Since $XS_1^{\text{lex}} \subset (XS_1)^{\text{lex}}$, for every j one has $x_j u_i = w_{p(i,j)}$ for some $1 \leq p(i,j) \leq s$. On the other hand, $x_j u_i v \in XS^{\text{xlex}} \subset (XS)^{\text{xlex}}$, therefore $w_{p(i,j)} v \in w_{p(i,j)} W_{p(i,j)}$, which implies that $v \in W_{p(i,j)}$. Hence $T_i \subset W_{p(i,j)}$, so that $T_i^{\text{lex}} \subset W_{p(i,j)}^{\text{lex}}$ for every i. This shows the inclusion $XS^{\text{bilex}} \subset (XS)^{\text{bilex}}$.

As at the beginning of the proof, one has $YT_i^{\text{lex}} \subset (YT_i)^{\text{lex}}$. Thus $YS^{\text{bilex}} = \bigcup_{i=1}^{l} u_i (YT_i^{\text{lex}}) \subset \bigcup_{i=1}^{l} u_i (YT_i)^{\text{lex}}$. On the other hand, since $YS^{\text{xlex}} = \bigcup_{i=1}^{l} u_i (YT_i)$, one obtains that $(YS^{\text{xlex}})^{\text{ylex}} = \bigcup_{i=1}^{l} u_i (YT_i)^{\text{lex}}$. Thus $YS^{\text{bilex}} \subset (YS^{\text{xlex}})^{\text{ylex}}$.

Now we show that $YS^{\text{xlex}} \subset (YS)^{\text{xlex}}$, and this will conclude the proof since by Lemma 4.11(b) one has that $(YS^{\text{xlex}})^{\text{ylex}} \subset ((YS)^{\text{xlex}})^{\text{ylex}} = (YS)^{\text{bilex}}$. Let $Y\{v_1, \ldots, v_k\} = \{z_1, \ldots, z_q\}$. Then we have the decomposition $YS = \bigcup_{i=1}^{q} U_i z_i$ with $U_i \subset M_{(b,0)}$. It is clear that for each j there is a $p(j)$, $1 \leq p(j) \leq q$, such that $S_j \subset U_{p(j)}$. Therefore $S_j^{\text{lex}} \subset U_{p(j)}^{\text{lex}}$. Since $(YS)^{\text{xlex}} = \bigcup_{i=1}^{q} U_i^{\text{lex}} z_i$, one obtains that $YS^{\text{xlex}} \subset (YS)^{\text{xlex}}$. □

Lemma 4.13. *Let $I \subset R$ be a strongly bistable ideal. Then there exists a bilex ideal I^{bilex} with the same bigraded Hilbert function.*

Proof. We have $I = \bigoplus I_{(b,d)}$ and each bigraded component $I_{(b,d)}$ is spanned by a strongly bistable set $S_{(b,d)}$ in $M_{(b,d)}$. For each (b,d), let $I_{(b,d)}^{\text{bilex}}$ be the vector space spanned by the bilex set $(S_{(b,d)})^{\text{bilex}}$ defined in Theorem 4.8. Set $I^{\text{bilex}} = \bigoplus I_{(b,d)}^{\text{bilex}}$. It remains to show that I^{bilex} is an ideal. By Lemma 4.12 for every (b,d) one has $XI_{(b,d)}^{\text{bilex}} \subset (XI_{(b,d)})^{\text{bilex}}$. Since $XI_{(b,d)} \subset I_{(b+1,d)}$, by Lemma 4.11 one obtains that $(XI_{(b,d)})^{\text{bilex}} \subset I_{(b+1,d)}^{\text{bilex}}$. Therefore $XI_{(b,d)}^{\text{bilex}} \subset I_{(b+1,d)}^{\text{bilex}}$. Similarly one has $YI_{(b,d)}^{\text{bilex}} \subset I_{(b,d+1)}^{\text{bilex}}$. This proves that I^{bilex} is an ideal. □

Now we are able to prove:

Theorem 4.14. *Let $I \subset R$ be a bigraded ideal. Then there exists a bilex ideal I^{bilex} with the same bigraded Hilbert function.*

Proof. Since any initial ideal $\text{in}(I)$ has the same bigraded Hilbert function as I, we can assume that I is a monomial ideal. Since the Hilbert function of a monomial ideal does not depend on the ground field, we can assume chara $k = 0$. Then $\text{bigin}(I)$ is strongly bistable (1.6). Now the claim of the theorem follows from Lemma 4.13. □

The following example shows that a bilex ideal with the same bigraded Hilbert function as I is not unique.

Example 4.15. Let S be the set defined in Example 4.10 and let I be the ideal generated by S. Using same notation as in the proof of Lemma 4.13, let $I_{(b,d)}^{\text{bilex}_2}$ be the vector

space spanned by the bilex set $((S_{(b,d)})^{\mathrm{ylex}})^{\mathrm{xlex}}$, and set $I^{\mathrm{bilex}_2} = \oplus I^{\mathrm{bilex}_2}_{(b,d)}$. Then I^{bilex_2} and I^{bilex} are two different bilex ideals with the same bigraded Hilbert function as I. Moreover,

$$I^{\mathrm{bilex}} = (S^{\mathrm{bilex}}, x_2^3 y_1 y_3); \quad I^{\mathrm{bilex}_2} = (S^{\mathrm{bilex}_2}, x_1 x_3 y_2^3)$$

so that they are generated in different bidegrees.

We recall the definition of the Macaulay expansion. Let a and i be two positive integers. Then a can be written uniquely in the form

$$a = \binom{a_i}{i} + \binom{a_{i-1}}{i-1} + \cdots + \binom{a_j}{j},$$

where $a_i > a_{i-1} > \cdots > a_j \geq j \geq 1$, see e.g. [5, Chapter 4]. Set

$$a^{\langle i \rangle} = \binom{a_i + 1}{i+1} + \binom{a_{i-1} + 1}{i} + \cdots + \binom{a_j + 1}{j+1}.$$

Proposition 4.16. *Let $L \subset M_{(b,d)}$ be a lexsegment. Denote by $\langle L \rangle$ the k-vector space spanned by L. Let*

$$\dim_k R_{(b,d)}/\langle L \rangle = \binom{m+d-1}{d} q + r$$

be the Euclidean division of $\dim_k R_{(b,d)}/\langle L \rangle$ by $\binom{m+d-1}{d}$. Then

$$\dim_k R_{(b,d+1)}/R_{(0,1)}\langle L \rangle = \binom{m+d}{d+1} q + r^{\langle d \rangle}.$$

Proof. Let $L = L(u)$ and write $u = u_x u_y$ with $u_x \in M_{(b,0)}$, $u_y \in M_{(0,d)}$. First we show

$$L(u) = \left(\bigcup_{\tilde{u} \in L(u_x) \setminus \{u_x\}} \tilde{u} M_{(0,d)} \right) \cup u_x L(u_y). \tag{5}$$

By definition of a lexsegment and in view of the considered monomial order, it follows that $u_x L(u_y) \subset L(u)$ and $(L(u_x) \setminus \{u_x\}) M_{(0,d)} \subset L(u)$. On the other hand, for every $w \in L(u)$, decomposing $w = w_x w_y$, one has $w_x \geq u_x$. If $w_x > u_x$, then $w \in (L(u_x) \setminus \{u_x\}) M_{(0,d)}$, and if $w_x = u_x$, then $w_y \geq u_y$, so that $w \in u_x L(u_y)$. This shows the desired decomposition (5).

For a monomial $w \in M_{(b,d)}$, set $E(w) = \{z \in M_{(b,d)} : z < w\}$. By (5) one obtains

$$E(u) = \left(\bigcup_{\tilde{u} \in E(u_x)} \tilde{u} M_{(0,d)} \right) \cup u_x E(u_y).$$

Therefore denoting $q = |E(u_x)|$ and $r = |E(u_y)|$, we get $|E(u)| = |M_{(0,d)}| q + r = \binom{m+d-1}{d} q + r$. Since $0 \leq r < \binom{m+d-1}{d}$, the integers q and r are defined uniquely by $|E(u)|$ and $\binom{m+d-1}{d}$.

By Lemma 4.6(b) we have $YL(u) = L(uy_m)$. On the other hand, from decomposition (5) we get

$$YL(u) = \left(\bigcup_{\tilde{u} \in L(u_x) \setminus \{u_x\}} \tilde{u} M_{(0,d+1)} \right) \cup u_x YL(u_y).$$

Since $YL(u_y) = L(u_y y_m)$, one obtains $E(uy_m) = E(u_x)M_{(0,d+1)} \cup u_x E(u_y y_m)$. Hence $\dim_k R_{(b,d+1)}/R_{(0,1)}\langle L \rangle = |E(uy_m)| = \binom{d+m}{d+1} q + |E(u_y y_m)|$. By Macaulay's Theorem we have $|E(u_y y_m)| = r^{\langle d \rangle}$ (see e.g. [5, Proposition 4.2.8]), and this concludes the proof. □

The proof of the following inequality uses exactly the same arguments as those of Bigatti's Theorem [4] (cf. Remark 4.2(a)). For convenience of the reader, we give the main steps.

Lemma 4.17. *Let $S \subset M_{(b,d)}$ be a strongly bistable set, and let $L \subset M_{(b,d)}$ be a lexsegment with $|L| \leq |S|$. Then*

$$m_{y,\leq i}(L) \leq m_{y,\leq i}(S) \quad \textit{for all } i.$$

Proof. We proceed by induction on m. The case $m = 1$ being clear, assume $m > 1$. The subsets S and L may be uniquely decomposed:

$$S = \bigcup_{j=0}^{d} S_j y_m^j, \quad L = \bigcup_{j=0}^{d} L_j y_m^j$$

with $S_j, L_j \subset M_{(b,d-j)} \cap k[x_1, \ldots, x_n, y_1, \ldots, y_{m-1}]$. It is easy to check that for every j, S_j is a strongly bistable subset of $k[x_1, \ldots, x_n, y_1, \ldots, y_{m-1}]$. One has

$$m_{y,\leq m}(L) = |L| \leq |S| = m_{y,\leq m}(S)$$

and

$$m_{y,\leq m-1}(L) \leq m_{y,\leq m-1}(S) \quad \Leftrightarrow \quad |L_0| \leq |S_0|.$$

Define

$$S^* = \bigcup_{j=0}^{d} S_j^{\text{lex}} y_m^j,$$

where S_j^{lex} is the lexsegment in $M_{(b,d-j)} \cap k[x_1, \ldots, x_n, y_1, \ldots, y_{m-1}]$ with $|S_j^{\text{lex}}| = |S_j|$ for all j.

We claim that S^* is strongly bistable. In fact, since all S_j^{lex} are lexsegments and in particular strongly bistable, it remains to show that for all j, denoting $Y_{m-1} = \{y_1, \ldots, y_{m-1}\}$, one has $Y_{m-1} S_j^{\text{lex}} \subset S_{j-1}^{\text{lex}}$. The set S being strongly bistable, it follows that $Y_{m-1} S_j \subset S_{j-1}$. By induction we have

$$m_{y,\leq i}(S_j^{\text{lex}}) \leq m_{y,\leq i}(S_j) \quad \text{for all } 1 \leq i \leq m-1.$$

Thus, applying Lemma 4.3(c), we get

$$|Y_{m-1}S_j^{\mathrm{lex}}| \le |Y_{m-1}S_j| \le |S_{j-1}| = |S_{j-1}^{\mathrm{lex}}|.$$

Since by Remark 4.2(b) $Y_{m-1}S_j^{\mathrm{lex}}$ is a lexsegment, this implies that $Y_{m-1}S_j^{\mathrm{lex}} \subset S_{j-1}^{\mathrm{lex}}$.

Replacing S by S^*, we may assume that all S_j are lexsegments.

For a subset $T \subset M_{(b,d)}$ set $\min(T) =$ the least monomial in T. For a monomial $u = x^\alpha y_1^{a_1} \dots y_m^{a_m}$, define $\hat{u} = u$ if $a_m = 0$, and $\hat{u} = u(y_{m-1}/y_m)^{a_m}$ if $a_m > 0$. Then, as in [4], one shows:

(a) $u, v \in M_{(b,d)}$, $u \le v \Rightarrow \hat{u} \le \hat{v}$;

(b) for every strongly bistable set S, one has $\widehat{\min}(S) = \min(S_0)$.

We claim that $\min(L) \ge \min(S)$. Indeed, assuming $\min(L) < \min(S)$, since $L = L(\min(L))$, we get $S \subset L$ and $S \ne L$ and from here the contradiction $|S| < |L|$. Now, by (a) and (b), we obtain

$$\min(L_0) = \widehat{\min}(L) \ge \widehat{\min}(S) = \min(S_0).$$

Since L_0 and S_0 are both lexsegments in $M_{(b,d)}$, the last inequality implies that $L_0 \subset S_0$. Thus $|L_0| \le |S_0|$. Moreover, we have by induction

$$m_{y,\le i}(L_0) \le m_{y,\le i}(S_0) \quad \text{for } 1 \le i \le m-1.$$

Since for $i < m-1$ one has $m_{y,\le i}(L) = m_{y,\le i}(L_0)$ and $m_{y,\le i}(S) = m_{y,\le i}(S_0)$, this concludes the proof of the lemma. □

Next we give necessary conditions for the bigraded Hilbert function. We noted that a similar formula was found in [3] in the study of Hilbert functions of graded algebras. It seems to us that this result does not imply our theorem below.

Theorem 4.18. *Let* $I = \bigoplus I_{(b,d)} \subset R$ *be a bigraded ideal, and let* $h(b,d) = \dim_k R_{(b,d)}/I_{(b,d)}$. *Moreover, let*

$$h(b,d) = \binom{m+d-1}{d} q + r \quad \text{and} \quad h(b,d) = \binom{n+b-1}{b} q_1 + r_1$$

be, respectively, the Euclidean division of $h(b,d)$ *by* $\binom{m+d-1}{d}$ *and by* $\binom{n+b-1}{b}$. *Then*

(a) $h(b,d+1) \le \binom{m+d}{d+1} q + r^{\langle d \rangle}$;

(b) $h(b+1,d) \le \binom{n+b}{b+1} q_1 + r_1^{\langle b \rangle}$.

Proof. Arguing as in the proof of Theorem 4.14 we can assume that I is a strongly bistable ideal.

Fix an integer $b \ge 0$ and consider $I_b = \bigoplus_{d \ge 0} I_{(b,d)} \subset R_b = \bigoplus_{d \ge 0} R_{(b,d)}$ as graded modules over $A = k[y_1, \dots, y_m]$. For every d, let $(I_{(b,d)})^{\mathrm{lex}}$ be the vector space spanned by the lexsegment consisting of the first $\dim_k I_{(b,d)}$ monomials in $M_{(b,d)}$. Set $I_b^{\mathrm{lex}} = \sum_{d \ge 0} (I_{(b,d)})^{\mathrm{lex}}$.

Applying Lemmas 4.3(c) and 4.17 we obtain the inequality

$$\dim_k R_{(0,1)}(I_{(b,d)})^{\mathrm{lex}} \le \dim_k R_{(0,1)} I_{(b,d)}.$$

Since $R_{(0,1)}I_{(b,d)} \subset I_{(b,d+1)}$, we get

$$\dim_k R_{(0,1)}(I_{(b,d)})^{\mathrm{lex}} \leq \dim_k I_{(b,d+1)} = \dim_k (I_{(b,d+1)})^{\mathrm{lex}}.$$

According to Lemma 4.6(b), $R_{(0,1)}(I_{(b,d)})^{\mathrm{lex}}$ is a vector space spanned by a lexsegment. Thus $R_{(0,1)}(I_{(b,d)})^{\mathrm{lex}} \subset (I_{(b,d+1)})^{\mathrm{lex}}$. This shows that I_b^{lex} is a graded module over A. Moreover, it has the same Hilbert function as I_b, i.e. $\dim_k R_{(b,d)}/(I_b^{\mathrm{lex}})_d = h(b,d)$ for every d. Then one obtains

$$\begin{aligned} h(b,d+1) &= \dim_k R_{(b,d+1)}/(I_b^{\mathrm{lex}})_{d+1} \leq \dim_k R_{(b,d+1)}/R_{(0,1)}(I_{(b,d)})^{\mathrm{lex}} \\ &= \binom{m+d}{d+1} q + r^{\langle d \rangle}. \end{aligned}$$

The last equality follows from Proposition 4.16. This shows (a).

Similarly, one proves (b) by considering the degree lexicographical order defined by $y_1 > \cdots > y_m > x_1 > \cdots > x_n$ (cf. Remark 4.7(b)). □

The following simple example shows that conditions (a) and (b) of Theorem 4.18 are not sufficient for a function to be the bigraded Hilbert function of a bigraded algebra.

Example 4.19. Let $h(1,0) = n = 3$, $h(0,1) = m = 2$, $h(1,1) = 2$, $h(2,0) = 6$, $h(0,2) = 3$, $h(2,1) = 3$, $h(1,2) = 3$ and $h(b,d) = 0$ if both $b > 1$ and $d > 1$. Then the only bilex sets S in $M_{(1,1)} \subset k[x_1, x_2, x_3, y_1, y_2]$ such that $|S| = |M_{(1,1)}| - h(1,1) = 4$ are:

$$S_1 = \{x_1y_1, x_1y_2, x_2y_1, x_3y_1\}, \quad S_2 = \{x_1y_1, x_1y_2, x_2y_1, x_2y_2\}.$$

However $|XS_1| = 9$, $|YS_1| = 7 > |M_{(1,2)}| - h(1,2)$, and $|XS_2| = 10 > |M_{(2,1)}| - h(2,1)$, $|YS_2| = 6$, so that there is no bilex ideal, and hence by Theorem 4.14 no bigraded ideal whose bigraded Hilbert function coincides with the given function.

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